

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Optical Sorter Faulting Mid-Shift

Who this is for and what it does: a material recovery facility (MRF) maintenance tech or plant engineer chasing an optical/near-infrared (NIR) sorter that runs fine, then faults out two or three times a shift. This walks you from symptom to likely cause to what to check and log, so you can find the real trigger instead of resetting and hoping.

How to use this

Keep it near the sorter panel. When the machine faults, do not just clear it and walk away. Note the time, what the fault said, and what the line was doing in the thirty seconds before. The pattern in those notes is usually the answer. Work top to bottom the first time; after that, jump to the section that matches your fault code. Nothing here asks you to bypass a guard, an interlock, or a protective trip. If a fix would require that, it is the wrong fix.

A quick note on terms. An optical sorter here means a belt or free-fall unit with an NIR (near-infrared) sensor that identifies material by how it reflects light, and a bank of fast air-blast ejector valves that knock the targeted items off the stream. When people say "the machine faulted," they usually mean one of three families: an air/pneumatic fault, a sensor or detection fault, or a drive/motion fault. The cause is frequently upstream of where the fault shows up.

Start here: what was the line doing when it tripped?

The single most useful habit is to stop treating the sorter as one machine and start watching it as the last stage of a line. A sorter faults because something changed in what reached it or what fed it. Before you open the sorter cabinet, look up the line.

- Did a surge of material hit right before the fault? Watch the infeed for a slug, a clump that came off a screen or a pile that a loader just dropped on the infeed conveyor.
- Did upstream equipment cycle? A baler door, a screen clearing a wrap, a second sorter on the same air header firing hard.
- Did the fault land at the same clock time each day, or the same point in the material flow? Same clock time points to something schedule- or environment-driven: ambient heat cooking the compressor as the day warms, sun angle washing the optics, a scheduled upstream event, shift-change loading. Same material condition — a dense band or a surging feed — points to the

material and the demand it creates. Air-pressure sag lives on the material side: a dense band of target fires many valves at once and pulls the header down, so it tracks the feed, not the clock.

Log those three things every fault for one shift and the cause tends to separate itself out.

Symptom to cause to check

Air / pneumatic faults (low pressure, valve fault, eject fault)

This is the one crews miss most. Each ejector valve dumps a short blast of compressed air. When the sensor sees a dense band of target material, the controller can command dozens of valves in the same few milliseconds. That simultaneous demand pulls the header pressure down faster than the receiver and compressor can refill it. The sorter watches its own supply pressure, sees it sag below the minimum, and faults to protect ejection accuracy. The gauge you glance at reads fine two seconds later, so it looks like a phantom.

Likely cause	What to check	What to log
Pressure sag when many valves fire at once	Put a pressure gauge or data logger at the sorter's air inlet , not at the compressor. Watch it during a heavy target surge.	Lowest pressure seen and whether the fault time matches the dip.
Undersized or waterlogged receiver tank	Tank volume, and whether the automatic drain is actually clearing condensate. A wet or small receiver has no reserve for peak demand.	Tank pressure recovery time after a burst.
Stuck or leaking ejector valve	A valve that hangs partly open bleeds the header continuously and also fires wrong or drags on the object. It is a sort-quality fault and a hidden driver of header sag at the same time. Listen and feel along the manifold with the line idle for a valve that never fully seats.	Which valve/bank; whether header pressure creeps down even with no target present.
Compressor not keeping up / duty cycle maxed	Is the compressor loading and unloading normally, or running flat out? Check aftercooler and dryer.	Compressor cut-in/cut-out pressures; discharge temp.
Line pressure drop: long run, undersized pipe, clogged filter/regulator, quick-disconnects	Compare pressure at the compressor vs. at the sorter under load. A big gap under load is restriction.	Static pressure and pressure under peak eject at both ends.
Water in the air fouling valves	Bowl on the filter/separator; cycling of the dryer. Wet air sticks valves and drops eject force.	Water present at the bowl; ambient humidity that day.

The tell that separates a supply problem from a valve problem: if pressure at the inlet holds steady but a specific fault points at a valve or bank, it is likely the valve or its wiring. If inlet pressure dips with the fault, it is supply (receiver, compressor, or piping) even though the fault showed up on the sorter.

Numbers to check against your own machine's spec plate and data sheet (these are typical rules of thumb, not setpoints to dial in blindly): supply to MRF optical/NIR sorters is commonly around 85–100 psi (roughly 6–7 bar) at the sorter inlet. Under peak ejection the inlet should not sag below the manufacturer's stated minimum — often somewhere near 70–80 psi; below that, ejection force falls off and accuracy goes with it. A well-fed system holds within a few psi through a burst; a brief droop of more than about 10–15 psi, or any dip that crosses the sorter's own low-pressure trip, is too much and points at receiver/compressor/piping. For air quality, ISO 8573-1 is the reference for how much water and particulate is acceptable — wet air both fouls and drops eject force.

One instrument caveat, because it decides whether you see the fault at all: the sag you are chasing is sub-second. A standard needle gauge damps it out and reads "fine" two seconds later, hiding the very thing you are looking for. Use a fast/peak-hold gauge or a data logger sampling at least a few Hz at the sorter inlet, and watch it through an actual heavy target burst.

Do not raise the fault threshold to make the alarm quiet. That threshold protects sort quality and keeps you from firing valves on air that cannot actually eject. Fix the supply.

Sensor / detection faults (calibration, lamp, no-read, dirty window)

Likely cause	What to check	What to log
Fouled sensor window	Look at the glass under a light. Fines, dust film, condensation, or a fingerprint smear all cut the signal. Clean per the manufacturer's method — usually a soft cloth and approved cleaner, not a shop rag.	Time since last clean; how fast it re-fouls.
Failing or aged illumination lamps/LEDs	Lamp hours against rated life. NIR lamps dim before they die and detection drifts as they do.	Lamp hours; whether faults cluster late in lamp life.
Background/reference calibration drift	Run the built-in calibration or background reference routine. A worn or dirty background reference reads like a detection fault.	Calibration pass/fail and any drift trend.
Ambient light or heat on the optics	Sun through a roof vent, a work light, or heat haze off warm material can wash the read.	Does the fault track time of day or sun angle?

Likely cause	What to check	What to log
Steam or dust cloud in the sensing zone	Watch for a dust plume off a dry, fast belt or steam off warm wet material crossing the read.	Correlation with dry days or a specific material.

Windows foul faster than people expect in an MRF. If yours needs cleaning more than once a shift, that is data — it points to a dust source upstream or an air-knife/blow-off that has stopped clearing the glass, not to a bad sensor.

Feed presentation and belt faults (overload, tracking, speed mismatch)

An NIR sensor reads a surface. If material is piled deep, tumbling, or bunched, the sensor cannot see individual items and the ejectors fire at the wrong spots. That shows up as detection faults, eject faults, or a downstream quality alarm, and it is really a feed problem.

Likely cause	What to check	What to log
Burden depth too high (material piled, not a monolayer)	Watch the belt at the sensor. You want a near-monolayer — material spread roughly one item deep so the sensor sees individual pieces. If items are stacked or shingled two-plus deep across much of the belt, presentation is the problem, not the sensor.	Belt loading during good runs vs. faults.
Surges from upstream, not steady metering	Is the metering/accelerator belt or vibratory feeder delivering an even stream, or slugs?	Whether faults follow visible surges.
Belt speed-feedback signal lost or drifted	Eject timing is slaved to the belt encoder/tach, so eject position tracks belt speed automatically — as long as the sorter is actually reading speed. The failure is a bad, lost, or noisy feedback signal, or a belt speed changed after commissioning without re-teaching the sorter — not raw speed itself. Check the encoder/tach and its wiring, not just the VFD setpoint.	Encoder/tach signal quality; whether belt speed still matches what the sorter was commissioned with.
Trigger photo-eye / registration drift	On units that use a trigger eye plus encoder counts to place the eject, a dirty, misaligned, or failing photo-eye or a slipping encoder makes the puff land just behind or ahead of the object. Sort quality drifts with nothing "faulted." Check the trigger eye's aim and cleanliness and the encoder coupling.	Whether ejects land consistently off-position; time since the eye was last checked/cleaned.
Belt mistracking or worn/torn belt	Belt edges, lacing, and tracking. A wandering belt changes where material sits under the sensor.	Tracking offset; belt condition.

Likely cause	What to check	What to log
Variable-frequency drive (VFD) faults on the belt motor	Drive fault log for overcurrent, overtemp, or overload. A dragging or overloaded belt trips the drive.	Exact drive fault code and load at trip.

Feed and air interact. A surge overloads the belt *and* commands a burst of valves at the same instant, so you can get a feed fault and an air fault from one slug of material. Watching the infeed tells you which came first.

10-minute triage (run this before you call anyone)

- ☐ Note the exact fault text/code and the time. Photograph the HMI fault screen and the fault history list.
- ☐ Watch the infeed for one full fault cycle: is there a surge, a pile, or a clump right before it trips?
- ☐ Read the air pressure **at the sorter inlet** during a heavy target burst, using a fast/peak-hold gauge or a logger sampling at least a few Hz — a plain needle gauge will read fine and hide a sub-second dip. Does it dip when the fault hits?
- ☐ Check the filter/separator bowl for water, and confirm the receiver's automatic drain is working and blow down any accumulated condensate. Do not drain the tank itself mid-shift — that dumps the air reserve you are trying to diagnose.
- ☐ Look at the sensor window under a light; clean it if it is filmed and note how dirty it was.
- ☐ Confirm belt speed and loading look like a spread-out layer, not a heap.
- ☐ Pull the VFD fault log and the sorter's own fault history; look for a repeating code or a repeating time.
- ☐ Enable the PLC's fault-history/sequence-of-events (SOE) buffer if it is not already on, then read the cascade backward to the first-out event — the trigger, not the last alarm to latch.
- ☐ Check whether a second machine on the same air header fires at the same moment.
- ☐ Write down: same clock time, or same material condition? Clock time points at schedule/environment (heat, sun, a timed upstream event); material condition points at the feed and the air demand it creates.

If the triage points clearly at air supply or feed and you can address it without touching a protective function, do it and watch for a shift. If it points into the drive, the sensor internals, or the control logic, gather the logs above before opening anything.

A word on safety

The hazard specific to an optical sorter is the one you cannot see: the manifold and air lines hold pressure after the machine stops. Before you open a cabinet, reach near the belt, or break into a pneumatic line, follow your site's lockout/tagout and bleed the stored air down to zero — a charged ejector bank can fire and a loaded fitting can let go. And keep the diagnostic honest: if the sorter faults because its air supply cannot support accurate ejection, the fix is the air supply, not a higher low-pressure threshold that lets it fire on pressure it does not have. Do not raise, jumper, or defeat a low-pressure trip, guard interlock, e-stop, or light curtain to keep the line moving.

Honest limits

This guide covers the causes I see repeat across MRF optical sorters regardless of brand: air supply behavior under peak demand, window and lighting condition, feed presentation, and belt/drive faults. It cannot tell you the meaning of a specific manufacturer's fault code, and it is not a substitute for the machine's own manual or a controls drawing. Some faults live in the controller program, a network dropout between sensor and valve driver, a failing valve bank, or a firmware quirk, and those need instrument-level work with the right tools. If your logging shows the trigger is none of the above, that itself is useful: it means the next step is inside the control system, and you will know what to point a specialist at.

If this keeps getting handed between vendors

The frustrating version of this problem is the one where the sorter vendor says the air is fine, the compressor vendor says the sorter is oversensitive, and the fault keeps happening. That standoff usually breaks the moment someone logs pressure at the sorter inlet and material at the infeed at the same time, on the same clock, and lets the data say which one moved first. If you have been around that loop a couple of times and want a vendor-neutral set of eyes on the whole line rather than the one machine, I am happy to talk it through: struleautomation.com or info@struleautomation.com

— Jonathan Gilmour, Strule Automation